

SwiftRoute Logistics — Data Preparation & QA Workbook

Purpose	This workbook is the first stop for the raw data extracts — the same role Excel plays in almost every real analytics team before data is loaded into a database. It validates the four source files, documents structure and known data-quality items, and produces a first pass of business-question answers using native Excel formulas (SUMIFS/COUNTIFS/AVERAGEIFS) — the kind of quick turnaround a stakeholder expects before a full SQL/BI build exists.
Source files	Drivers.csv (55 rows), Hubs.csv (6 rows), Orders.csv (27,979 rows), Vehicles.csv (45 rows). Original files were UTF-16 tab-delimited exports; re-saved here as standard tables.
How this fits the project	1) Excel — clean, validate, first-pass answers (this file). 2) SQLite/SQL notebook — operational queries needing joins, ranking, trend analysis (SwiftRoute_SQL_Analysis.ipynb). 3) Python notebook — statistical significance testing and predictive modeling (SwiftRoute_Python_Analysis.ipynb). 4) Power BI — the executive dashboard stakeholders see day to day (LOGISTICS_DASHBOARD.pbix).
Sheet guide	DQ_Checks: data quality summary across all 4 tables. Delay_Reason_Analysis, Cancellation_Analysis, Hub_Capacity_Utilization, Driver_Workload, Vehicle_Utilization: pivot-style business-question answers built with live formulas. Orders_Raw / Drivers_Raw / Hubs_Raw / Vehicles_Raw: the source data each analysis sheet references.
Data note	This is a synthetic dataset built for portfolio purposes. Several relationships in it are close to statistically flat (see DQ_Checks and the Python notebook) — this workbook reports that honestly rather than manufacturing findings that aren't in the data.

Data Quality Checks

Table	Check	Result	Flag
Orders	Row count	27979	
Orders	Duplicate Order IDs	0	OK
Orders	Missing Order Date	0	OK
Orders	Missing Delivery Time (expected for Cancelled)	252	
Orders	Cancelled orders (explains blank delivery time above)	252	
Orders	Delay Reason populated but Is Delayed = FALSE (should be 0)	0	OK
Orders	Is Delayed = TRUE but Delay Reason blank (should be 0)	0	OK
Orders	CSAT score out of expected 1-5 range	0	OK
Orders	Order Date range - earliest	2023-01-01	
Orders	Order Date range - latest	2024-12-31	
Orders	Driver IDs in Orders not found in Drivers table	0	OK
Orders	Vehicle codes in Orders not found in Vehicles table	0	OK
Orders	Hub names in Orders not found in Hubs table	0	OK
Drivers	Row count	55	
Drivers	Duplicate Driver IDs	0	OK
Drivers	Missing Hire Date	0	OK
Drivers	Performance Rating out of expected 1-5 range	0	OK
Vehicles	Row count	45	
Vehicles	Duplicate Vehicle Codes	0	OK
Vehicles	Vehicles with zero linked orders (see Vehicle_Utilization sheet)	4	
Hubs	Row count	6	
Orders↔Drivers	Orders where embedded 'Driver Name' text does NOT match the Drivers master table name for that Driver ID	27979	REVIEW

Notes

- Cancelled orders (252 total) legitimately have blank Delivery Time Hours and Actual Delivery Date — this is expected, not a data quality error.
- All Vehicle Name values in Orders match a Vehicle Code in the Vehicles table (0 unmatched) — the vehicles with zero linked orders are a genuine utilization finding, not a join/key mismatch. See Vehicle_Utilization sheet.
- All Driver IDs and Hub Names in Orders successfully match their master tables — referential integrity is clean across the dataset.
- The 'Driver Name' field inside Orders.csv is NOT reliably linked to Driver ID — always join Orders to Drivers on Driver ID/DriverID, never on the name text. This was caught during Excel QA and is why every downstream analysis (SQL notebook, Python notebook, this workbook's Driver_Workload sheet) joins on ID.

Delay Reason Analysis

Q: What are the top reasons orders are delayed, overall and by hub?

Delay Reason	Occurrences	% of All Delays
Customer Not Home	583	9.9%
Driver Unavailable	595	10.1%
Hub Processing Delay	589	10.0%
Incorrect Address	545	9.2%
Multiple Delivery Stops	586	9.9%
Package Sorting Error	603	10.2%
Road Construction	623	10.5%
Severe Weather	585	9.9%
Traffic Congestion	576	9.7%
Vehicle Breakdown	623	10.5%
Total Delayed Orders	5908	

Top Delay Reason by Hub

Hub Name	Top Delay Reason	Occurrences at this Hub
Austin Hub	Road Construction	99
Dallas Main Hub	Road Construction	178
El Paso Hub	Package Sorting Error	62
Fort Worth Hub	Package Sorting Error	80
Houston Hub	Vehicle Breakdown	172
San Antonio Hub	Traffic Congestion	102

Cancellation Analysis

Q: What is the cancellation rate, and does it vary by hub or vehicle type?

Overall Cancellation Rate **0.90%**

By Hub

Hub Name	Total Orders	Cancelled	Cancellation Rate
Austin Hub	4065	44	1.08%
Dallas Main Hub	7345	63	0.86%
El Paso Hub	2718	21	0.77%
Fort Worth Hub	3255	20	0.61%
Houston Hub	6875	68	0.99%
San Antonio Hub	3721	36	0.97%

By Vehicle Type

Vehicle Type	Total Orders	Cancelled	Cancellation Rate
Box Truck	1131	10	0.88%
Pickup	2837	28	0.99%
Truck	6531	62	0.95%
Van	17480	152	0.87%

Hub Capacity Utilization

Q: Which hubs run over their rated capacity, based on total 2-year delivered order volume?

Hub Name	Hub Capacity	Total Delivered Orders (2yr)	Avg Monthly Orders	Utilization vs Monthly Capacity
Austin Hub	220	4021	167.5	76.2%
Dallas Main Hub	250	7282	303.4	121.4%
El Paso Hub	150	2697	112.4	74.9%
Fort Worth Hub	180	3235	134.8	74.9%
Houston Hub	380	6807	283.6	74.6%
San Antonio Hub	200	3685	153.5	76.8%

Note: 'Avg Monthly Orders / Hub Capacity' approximates monthly utilization from the 2-year total. Dallas Main Hub is the only hub that runs consistently over 100% — see the SQL notebook for the full month-by-month trend (Question 3), which confirms this isn't a one-off spike.

Driver Workload Distribution

Driver Name	Driver ID	Deliveries (2yr)
Sarah Lopez	1	433
Lisa Davis	2	395
Steven Anderson	3	434
James Garcia	4	816
Christopher Martin	5	1281
Mary Moore	6	414
Steven Williams	7	421
Joseph Williams	8	411
Richard Brown	9	415
Christopher Williams	10	379
Matthew Anderson	11	1190
David Thompson	12	705
John Moore	13	374
Charles Harris	14	412
William Miller	15	668
Jessica Martinez	16	411
Daniel Martin	17	802
John Wilson	18	421
Sarah Martinez	19	429
Jennifer Martinez	20	788
Robert Davis	21	440
Charles Garcia	22	345
Elizabeth Davis	23	424
Matthew Williams	24	434
Daniel Davis	25	374
Mark Jones	26	442
Mark Brown	27	415
Barbara Clark	28	408
James Rodriguez	29	426
Jessica Lee	30	372
Matthew Wilson	31	413
Mary White	32	403
Lisa Miller	33	431
Christopher Miller	34	382
Kevin Miller	35	398
Karen Smith	36	410
Barbara Garcia	37	423
Daniel Lee	38	664
Patricia Miller	39	829

Joseph Thompson	40		461
Christopher Rodriguez	41		660
Susan Garcia	42		383
Thomas Thomas	43		1214
Susan Martinez	44		440
Mary Rodriguez	45		388
Jennifer Thompson	46		395
Charles Harris	47		440
Mark Harris	48		395
Daniel Martin	49		445
Joseph Rodriguez	50		426
Lisa Lee	51		420
Sarah Brown	52		384
Michael Johnson	53		412
William Brown	54		407
Michael Williams	55		425

Summary Statistics

Mean deliveries/driver	504.1
Std deviation	213.8
Max (busiest driver)	1281
Min (least busy driver)	345
Busiest : Least-busy ratio	3.7x
Busiest driver name	Christopher Martin

Vehicle Utilization

Q: Which vehicles are idle or under-utilized vs. overworked?

Vehicle Code	Vehicle Model	Vehicle Status	Orders Handled (Delivered)
FT-001	Mercedes Sprinter	Active	781
FT-002	Freightliner M2	Maintenance	675
FT-003	Mercedes Sprinter	Active	766
FT-004	Chevrolet Express	Active	315
FT-005	Chevrolet Silverado	Maintenance	1263
FT-006	Freightliner M2	Active	302
FT-007	Ford Transit	Active	709
FT-008	Freightliner M2	Active	801
FT-009	Isuzu NPR	Active	652
FT-010	Freightliner M2	Active	833
FT-011	Mercedes Sprinter	Active	304
FT-012	International DuraStar	Active	763
FT-013	Isuzu NPR	Active	842
FT-014	Mercedes Sprinter	Active	0
FT-015	Chevrolet Silverado	Active	819
FT-016	Mercedes Sprinter	Active	732
FT-017	Chevrolet Express	Active	802
FT-018	Ram ProMaster	Active	791
FT-019	Ford Transit	Active	743
FT-020	Freightliner M2	Maintenance	662
FT-021	International DuraStar	Active	786
FT-022	Ford Transit	Maintenance	783
FT-023	Chevrolet Express	Active	643
FT-024	Ford Transit	Maintenance	0
FT-025	Ram ProMaster	Maintenance	689
FT-026	Mercedes Sprinter	Active	809
FT-027	Ram ProMaster	Maintenance	1229
FT-028	Ford Transit	Active	726
FT-029	Chevrolet Express	Maintenance	295
FT-030	Ford F-150	Maintenance	653
FT-031	Mercedes Sprinter	Active	695
FT-032	Chevrolet Silverado	Active	779
FT-033	Ford Transit	Active	611
FT-034	International DuraStar	Active	652
FT-035	Freightliner M2	Maintenance	307
FT-036	Freightliner M2	Active	1193
FT-037	Freightliner M2	Active	294

FT-038	Ford F-150	Active		0
FT-039	Mercedes Sprinter	Active		644
FT-040	Ford F-150	Maintenance		666
FT-041	Freightliner M2	Active		279
FT-042	Isuzu NPR	Active		0
FT-043	International DuraStar	Maintenance		289
FT-044	International DuraStar	Active		838
FT-045	Mercedes Sprinter	Active		312

Summary

Vehicles with zero 4

Max orders handle 1263

Avg orders per vet 616.2